

Dashboard Storyboard

Hospital Operations & Patient Analytics Dashboard

Project Overview

Hospitals generate a vast amount of operational data every day, including patient admissions, appointments, medical records, staff information, bed allocations, and equipment usage. While this data is valuable, it is often stored across multiple systems, making it difficult for hospital administrators to gain a complete picture of hospital performance.

The goal of this project is to bring all of this information together into one interactive Power BI dashboard that provides meaningful insights into hospital operations. Instead of reviewing multiple reports manually, decision-makers can quickly understand patient flow, departmental performance, and resource utilization through a single integrated dashboard solution. This objective reflects the overall aim of the MedTrack_DV project described in the internship specification.

User Journey

The dashboard has been designed with hospital administrators and healthcare managers in mind. Rather than overwhelming users with every available metric at once, the report guides them through four connected dashboards, each focusing on a specific aspect of hospital operations.

The journey begins with a high-level overview of hospital performance. From there, users can explore patient movement, evaluate department performance, and finally understand how hospital resources are being utilized. Throughout the report, common filters such as Date Range, Department, and Gender allow users to investigate specific scenarios without losing context.

Dashboard 1 – Hospital Overview

The first dashboard serves as the executive summary of the hospital's performance. It answers the most important operational questions at a glance by presenting key indicators such as total patients, total admissions, revenue, average length of stay, occupancy rate, and bed utilization.

Trend charts help users understand how admissions, revenue, occupancy, and readmissions change over time, while additional visuals highlight appointment status, departmental revenue, and average length of stay.

This dashboard is intended to answer questions such as:

- How is the hospital performing overall?
- Are admissions increasing or decreasing?
- Which departments contribute the most revenue?
- Is occupancy remaining within expected levels?

By starting with these high-level metrics, hospital management can quickly assess the overall health of operations before exploring detailed analyses.

Dashboard 2 – Patient Flow Analytics

After reviewing the overall hospital performance, users move to the Patient Flow dashboard.

This page focuses on how patients move through the hospital—from admission to discharge—and whether that journey is efficient.

The dashboard presents metrics such as total admissions, total discharges, unique patients, average length of stay, and readmission rate. Visualizations compare admissions and discharges across departments while also showing patient demographics such as gender and age group.

These insights help answer questions like:

- Which departments admit the highest number of patients?
- Are discharge rates keeping pace with admissions?
- Which patient groups require more attention?
- Which departments experience higher readmission rates?

Understanding patient flow allows administrators to identify bottlenecks and improve operational efficiency.

Dashboard 3 – Department Analytics

The third dashboard shifts the focus from patients to departments.

Every department contributes differently to hospital operations. Some require more staff, while others depend heavily on specialized equipment. This dashboard compares departments based on staffing levels and equipment utilization.

Users can explore the distribution of doctors and nurses across departments while also monitoring equipment usage and departmental resource allocation.

The dashboard helps answer questions such as:

- Which departments have the highest staffing levels?
- Is equipment distributed appropriately?
- Which departments require additional resources?
- How balanced is the allocation of doctors and nurses?

These insights support workforce planning and departmental performance evaluation.

Dashboard 4 – Resource Utilization

The final dashboard focuses on the physical resources that keep the hospital running.

Hospital resources such as beds, rooms, wards, and medical equipment directly impact patient care. This dashboard provides visibility into how efficiently these resources are being utilized.

Users can view bed distribution across wards, room availability by department, equipment status, equipment types, monthly equipment usage, and maintenance activities.

These visualizations help hospital administrators answer questions like:

- Which wards contain the highest number of beds?
- Which departments use the most equipment?
- Is equipment being properly maintained?
- Are hospital resources being fully utilized?

This dashboard supports long-term planning and helps identify opportunities to improve operational efficiency.

Interactive Features

To make exploration easier, all four dashboards share the same filtering experience. Users can filter the report by date, department, or gender, and every visualization updates automatically. This allows managers to compare different departments, analyze trends over time, and investigate specific operational scenarios without navigating away from the report.

The dashboards are also designed with a consistent layout and color scheme so that users can move from one page to another without needing to relearn the interface.

Business Value

The dashboard transforms raw healthcare data into meaningful business insights that support faster and more informed decision-making.

With this solution, hospital management can:

- Monitor hospital performance in real time
- Improve patient flow
- Optimize departmental operations
- Better allocate staff and equipment
- Monitor resource utilization
- Support strategic planning with data-driven insights

By integrating multiple datasets into one interactive reporting solution, the dashboard reduces manual reporting effort while providing a comprehensive view of hospital operations.

Conclusion

The **Hospital Operations & Patient Analytics Dashboard** demonstrates how healthcare data can be transformed into an effective decision-support system using Microsoft Power BI. Through four connected dashboards, the solution provides a clear view of hospital performance, patient movement, departmental efficiency, and resource utilization. The interactive design allows users to move seamlessly from high-level summaries to detailed operational insights, making it a practical tool for hospital administrators and healthcare managers.